Paleomagnetism of Kamthi Sandstones of Upper Permian Age from Godavary Valley, India

R. K. VERMA AND M. S. BHALLA

National Geophysical Research Institute, Hyderabad, India

The paleomagnetism of fifty-seven rock samples of red sandstones, collected from three sites, belonging to the Kamthi formation of Upper Permian age has been studied. The natural remanent magnetization of most of the samples showed normal polarization with upward dip (declination close to 360° defined as normal). The intensity of magnetization of the samples ranged from 1 to 10 $\times$ 10^{-6} gauss. The samples were demagnetized in peak alternating fields of 50, 100, 200, 400, and 600 oersteds. The grouping of the vectors improved considerably on demagnetization. Mean direction for the three sites after ac cleaning was 306°E and −50° (up) after applying dip correction. The corresponding virtual geomagnetic pole is situated at 18°N, 53°W. The results are consistent with India's being in the southern hemisphere during Upper Permian times, when it might have been a part of Gondwanaland along with the Continents of Africa, Australia, Antarctica, and South America.

Introduction. The paleomagnetic studies of the Deccan traps (Upper Cretaceous to Miocene) and of the Rajmahal and Sylhet traps (probably Jurassic) provide evidence for the drift of the Indian land mass from southern latitudes since the Jurassic to its present position in the northern hemisphere [*Clegg et al.*, 1956, 1958; *Deutsch et al.*, 1958, 1959; *Athavale et al.*, 1963]. There is a wide gap from Jurassic to Precambrian in the study of paleomagneteism of Indian rocks. As continental drift is a major phenomenon, which appears to have occurred from the Carboniferous to Tertiary, it is important to study rock formations spanning this entire period of time.

Fine-grained red sandstones and shales have been found to be particularly suitable for paleomagnetic studies, as they have very often yielded stable directions of magnetization. Redbeds of Permian and Triassic age have been studied in other fragments of Gondwanaland. Results from Africa [*Opdyke*, 1964], South America [*Creer*, 1965], and Australia [*Irving*, 1963] are available. In this paper, results from fine-grained red sandstones of the Kamthi formation of Upper Permian age from Godavary Valley are reported.

Geology and sampling. The Wardha-Godavary valley is a well-known Gondwana trough in southern India, which runs in a general NW–SE direction from near Rajahmundry (17°N, 81°48′E) to near Chanda (20°N, 79°21′E). The valley is 50–65 km wide and is flanked on either side by Precambrian metamorphic rocks. It is filled with sediments ranging in age from Carboniferous to Cretaceous. The Kamthi formation is exposed over a distance of nearly 320 km along the western margin of the valley. The stratigraphic sequence [*King*, 1880; *Krishnan*, 1960] is given in Table 1.

The general strike of the formations is NW–SE, and the dip is about 10° to NE. It has been mentioned [*Pascoe*, 1959 p. 960] that the Kamthis, on account of their considerable thickness, may include part of the Panchet series, which are Triassic. In Godavary valley, however, there is no evidence of this combination. Near the sampling sites, the Kamthis overlie the Barakars of Lower Permian age, and further northwest they are overlain by Maleris of

TABLE 1. Stratigraphic Sequence of Rocks in Godavary Valley

Alluvium	Recent
Traps of Jangaom Valley	Upper Cretaceous
Chikiala sandstones	Middle Jurassic
Kota beds	
Maleri beds	Upper Triassic
Kamthi sandstones	Upper Permian
Barakar sandstones	Lower Permian
Talchir shales	Carboniferous
Sullavai sandstones	Lower Vidhyans or Lower Paleozoic
Pakhal series	Pre-Paleozoic
Gneisses	Precambrian

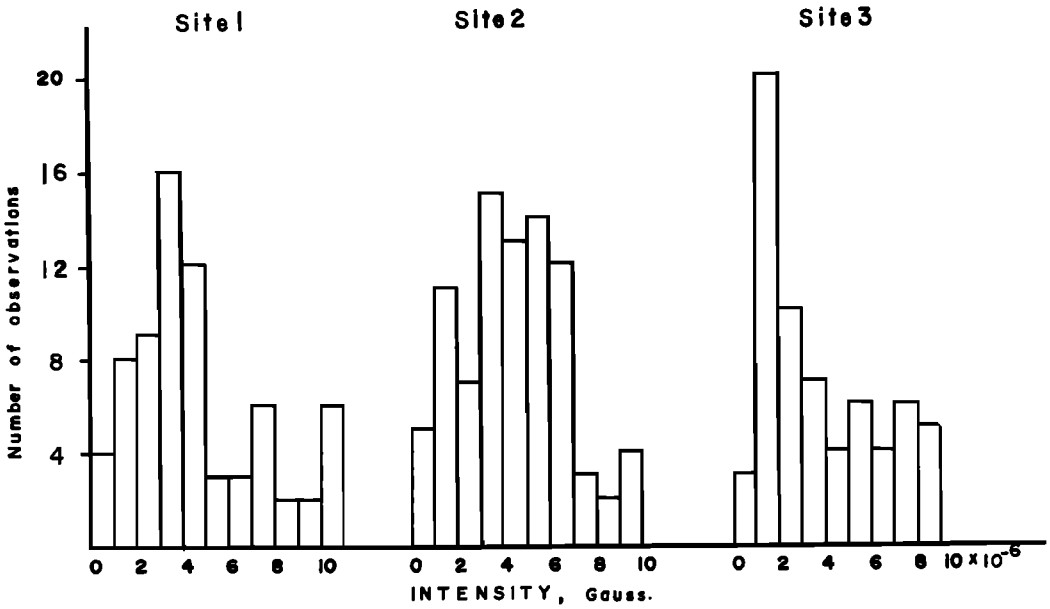


Fig. 1. Histogram of intensities of NRM for sites 1, 2, and 3.

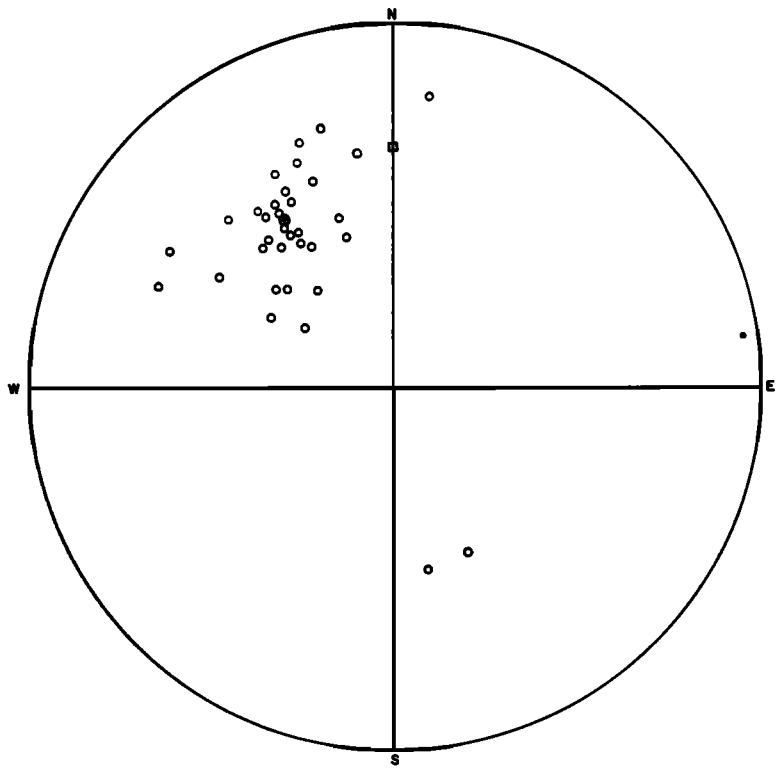


Fig. 2*a*. Equal area projection of NRM directions for sample from sites 1 and 2. Each sample represents a mean of several specimens. Downward inclinations are plotted as solid circle on the lower hemisphere and upward inclinations as open circles on the upper hemisphere. Up dips, ○; down dips, ●; present dipole field, □; mean direction, ⊕.

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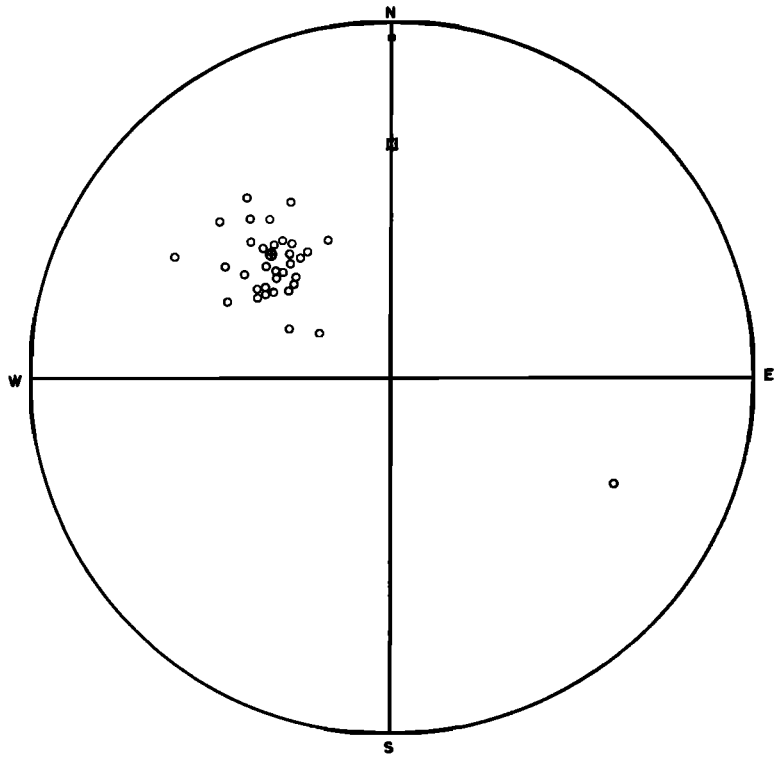


Fig. 2*b*. Directions of the remanent vector for sites 1 and 2 after magnetic cleaning in peak field of 600 oe. Equal area projection used. Symbols as in Figure 2*a*.

Upper Triassic age. The possibility that the uppermost Kamthis are of Lower Triassic age cannot be ruled out, although there is no paleontological evidence for this in the area of sampling. For the present, however, the age of the Kamthi beds has been taken to be the Upper Permian on the basis of presence of such fossils as *Equisetales, Phyllotheca indica, Pteridospermae, Glossopteris indica, Glossopteris communis, Glossopteris browniana, and*

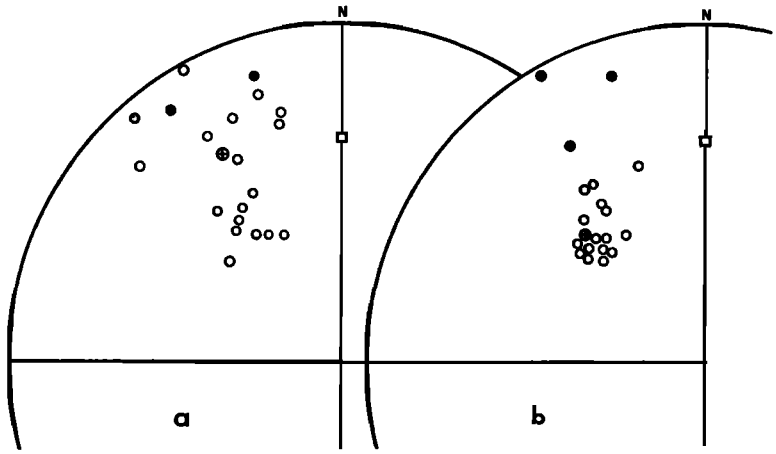


Fig. 3. Equal area projection of magnetic directions for site 3. (*a*) NRM directions; (*b*) after magnetic cleaning in 600-oe peak alternating fields. Symbols as in Figure 2*a*.

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TABLE 2. Variation of Precision k with Demagnetizing Field

Peak Field, oe	Site 1	Site 2	Site 3
0	7	110	18
400	50	220	80
600	65	220	160

Glossopteris Gangamopteris [*Pascoe*, 1959].

Fifty-seven oriented samples were collected from three separate quarries. Two quarries, sites 1 (14 samples) and 2 (23 samples), were situated near the village of Jaipuram (18°51′N, 79°37′E) and were about 2 km apart. The third quarry (site 3, 20 samples) was situated near the village of Rallapet (19°7′N, 79°35′E). At each site the samples were spread over a distance of about 200 meters and a thickness of 2 meters. The samples are fine-grained sandstones of reddish to purple color and have a concoidal fracture.

Results. Three or four cylinders of 2.5-cm diameter and about 2.0-cm length were cut from each sample. All measurements were made with an astatic magnetometer of sensitivity 3×10^{-7} oe/cm deflection at 4 meters optical path. At all sites the intensities of magnetization were found to vary, usually between 1 and 10×10^{-6} gauss. Mean intensities are for site 1, 3.8×10^{-6}; site 2, 4.3×10^{-6}; and site 3, 3.7×10^{-6}. A histogram of NRM intensities of all specimens is shown in Figure 1. The NRM directions of the remanent vector for each sample from sites 1 and 2 (combined) are plotted on equal area projection in Figure 2*a* and for site 3 in Figure 3*a*.

Most of the specimens showed normal magnetization with upward inclination. Normal magnetization is defined on the basis of declination, values close to 360° being defined as normal and values close to 180° as reverse. Two samples from site 1 had reverse polarization and upward dips, and one sample had a down-

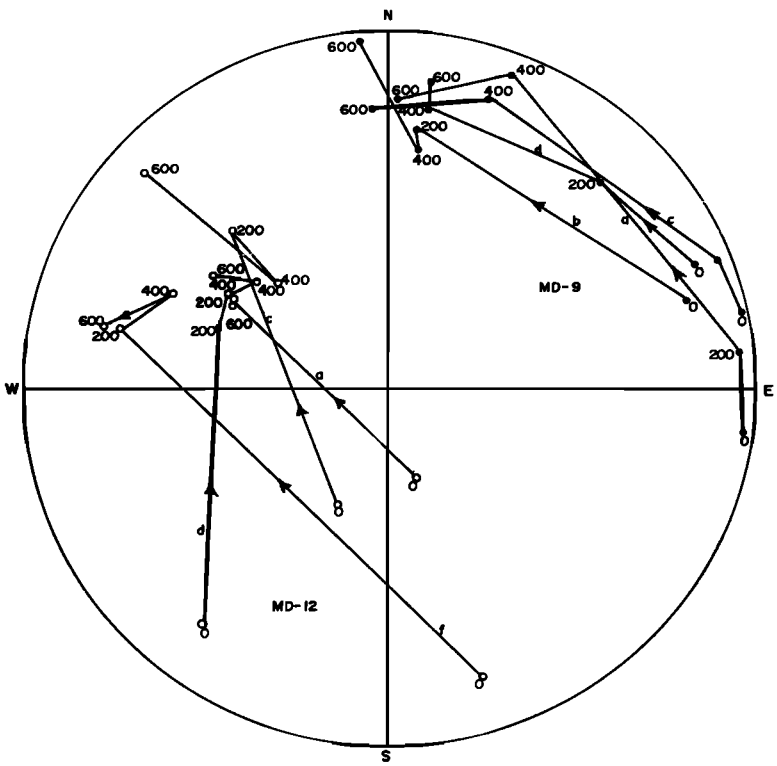


Fig. 4. Change in magnetic directions for eight specimens from two samples MD-12 and MD-9 from site 1 as a result of ac cleaning in peak fields of 200, 400, and 600 oe plotted on equal area projection.

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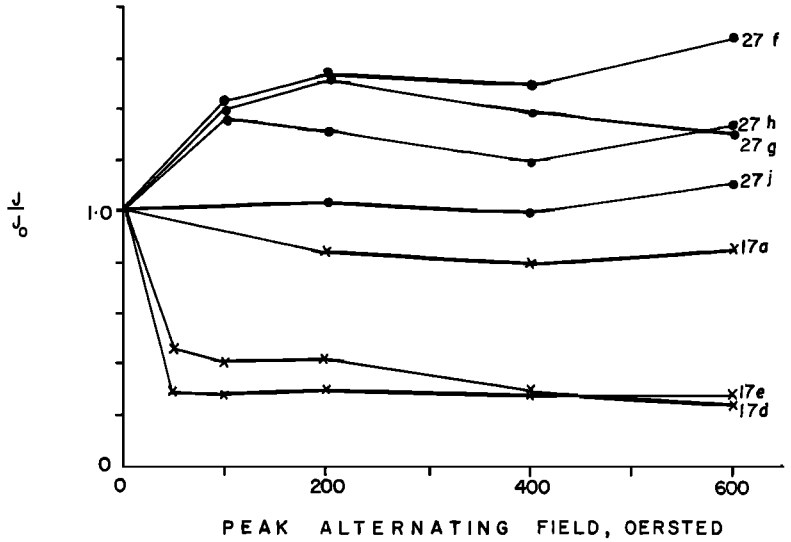


Fig. 5. Normalized ac demagnetization curves for specimens from two samples (MF27 and MF17) from site 2.

ward dip with a declination of 80°. From site 3, seventeen samples showed normal magnetization with upward dip, and three samples showed normal magnetization with downward dip.

Disks from each sample were demagnetized in successive peak alternating fields of 50, 100, 200, 400, and 600 oersteds in apparatus similar to that used by *Creer* [1959]. The grouping of the vectors was improved. *Fishers'* [1953] precision parameter k is given in Table 2 for before and after demagnetization in peak fields of 400 and 600 oe. The directions after magnetic cleaning in 600-oe peak field are shown in Figures 2*b* and 3*b*. Most of the directions remain in the northwest quadrant. A few samples showed considerable change in direction on cleaning, as is shown by samples MD-12 and MD-9 from site 1 (Figure 4).

From site 3, two samples retained a downward inclination despite ac cleaning, and one sample changed from upward to downward inclination. Typical ac demagnetization curves of normalized intensity for two rock samples (MF27 and MF17) from site 2 are shown in Figure 5. Similar curves for one sample (MG-4) from site 3 are shown in Figure 6. These samples showed very little changes in directions of magnetization as a result of ac cleaning. For most of the specimens there was a change in intensity of about 40% after cleaning in 600-oe peak field. Samples with both downward and upward dips showed similar behavior.

Most of rocks showed high stability. Thermal demagnetization of the isothermal remanent magnetization acquired in an 8000-oe field was carried out. A few samples with downward inclinations were found to have low Curie point (about 150°C) components. These samples were rejected in the calculation of mean direction.

Analysis and discussion. The mean direction for each site and other statistical parameters are given in Table 3. Virtual geomagnetic pole positions were calculated assuming an axial geocentric dipole field during the period of magnetization of the rocks. For all three sites the pole positions are fairly close. The over-all mean for the three sites is 306°E and −50°

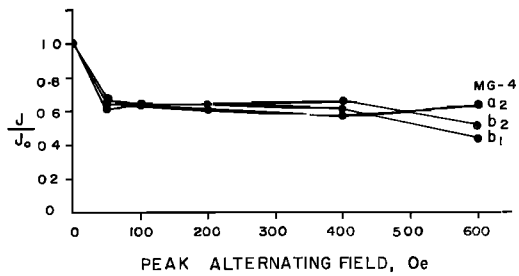


Fig. 6. Typical ac demagnetization curves of normalized intensity for one sample, MG-4, from site 3.

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TABLE 3. Site Parameters for Kamthi Sandstones of Upper Permian Age after ac Cleaning in 600-oe Field Strength

Locality	Site	N	E	n	D, °E	I, deg	D', °E	I', deg	k	R	α_{95}, deg	Pole Position Lat., °N	Pole Position Long., °W	Pole Position λ, °S
Jaipuram 18°52′N, 79°35′E	1	12	2	55	315	−44	303	−44	65	11 83	5	18	48	26
Jaipuram 18°52′N, 79°35′E	2	23	0	87	318	−53	306	−53	220	22.9	2	15	56	34
Rallapet 19°7′N, 79°30′E	3	17	3	65	321	−41	309	−47	160	16.9	3	22	52	28
	1+2+3	52	5	207	318	−50	306	−50	85	51.4	2	18	53	31

Notes

k, $(N - 1)/(N - R)$ is least estimated of Fisher's dispersion parameter.
R, length of the result of N unit vectors.
α_{95}, semiangle of the cone of 95% confidence.
N, number of samples included for analysis of pole position.
E, number excluded for calculation of mean position.
n, number of specimens (disk) studied.
D and I, declination and inclination, respectively.
D' and I', declination and inclination, respectively, after dip correction
λ, ancient latitude of site.

(upward inclination) after applying dip correction. The corresponding mean pole position is 18°N, 53°W. The ancient latitude for the sites is 31°S, and for Nagpur (which is a centrally located reference town for all paleolatitude values for data from India) it is 28°S. The results suggest that India was in the southern hemisphere during the Upper Permian times, when it could have been a part of Gondwanaland.

Results of paleomagnetic studies of Permian rocks from other parts of the world show that the geomagnetic field was reversed during the Permian times [*Irving*, 1964]. The present results indicate, however, a normal polarity. It appears therefore that there could have been a geomagnetic field reversal sometime toward the end of the Permian.

The virtual geomagnetic pole positions for the lower Deccan traps of Upper Cretaceous age from several different localities Amba, Linga, Khandala, Nipani [*Deutsch et al.*, 1958, 1959], Chinocholi [*Verma and Rao*, 1963] and those of the Rajmahal and Sylhet traps of Jurassic age [*Clegg et al.*, 1958, *Athavale et al.*, 1963] are compared with the pole position of the Kamthi beds in Figure 7. The present results imply a relatively small latitudinal displacement of the land mass of India with respect to the geographic pole between Upper Permian and Middle Jurassic times, as compared with the corresponding movement between Jurassic and Upper Cretaceous times. It is not likely that the directions of the Kamthis could have been influenced by any igneous activity of Jurassic age, because no such activity exists nearby.

The question of the breakup of the Gond-

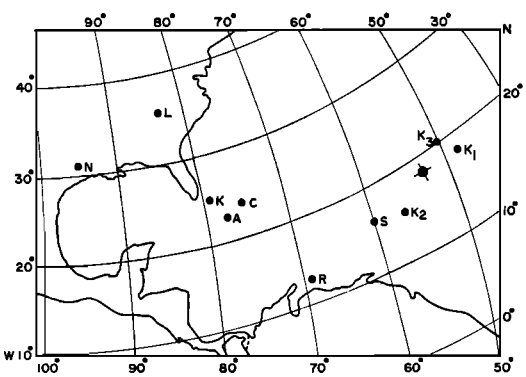


Fig. 7. Virtual geomagnetic pole positions for lower Deccan traps of Upper Cretaceous age from different localities: N, Nipani; L, Linga; K, Khandala; C, Chincholi; and A, Amba. R and S are pole positions for Rajmahal and Sylhet traps of Jurassic age. K_1, K_2, and K_3 are pole positions for the three sites in present studies. The circle with a cross through it represents mean pole for Kamthis.

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wanic continents from the Gondwanaland is of general interest [*Creer*, 1965]. The present results suggest that the breakup could have occurred sometime between Paleozoic and Mesozoic. More detailed studies of the Upper and the Lower Permian formations (including the Kamthi formation) from various parts of India are in progress. The present results may, however, be considered as tentative.

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